Cubic Pisot Floors: Composite Windows, Sparse Exceptions, and Finite-Sieve Barriers

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Revised research manuscript: residual fields and finite-prime survival

Abstract

We investigate Saito's question on composite values of $\lfloor \beta^{3^n} \rfloor$ for cubic Pisot numbers, using the supplied constant-race manuscripts as prior work. A constructive prime-divisor argument places every fixed window of shifted floors simultaneously on a composite arithmetic progression for fourteen coefficient classes modulo three. A separate finite-field theorem gives exactly p^{d-1} coefficient vectors at every shifted trace value for odd degree d and primes $p \equiv 2 \pmod 3$. Combined with geometric separation and the large sieve, it yields a uniform $\exp(-c\sqrt{\log m})$ exceptional proportion in coefficient boxes, for every $c < 1/(e\sqrt{2\log 3})$. We also strengthen a Pell-family exclusion by three exact modular certificates, refine conditional finite prime/composite patterns to prescribed exact local multiplicities, and count balanced Pisot units in fixed congruence cosets. A residual-field analysis excludes every nonzero rational power of Mills' constant from broad Galois and cyclotomic fields. Conversely, in every totally real cubic field admitting a three-adic embedding, we construct infinitely many Saito-balanced Pisot numbers avoiding any prescribed finite set of prime divisors at every odd exponent. Previously established resolutions and conditional prefix results are identified explicitly. Neither the full cubic Pisot problem, irrationality of the least square-clock constant, nor unconditional transcendence of Mills' constant is proved.

1 The problem, the source boundary, and the new results

Let

$$A = \min\{x > 1 : \lfloor x^{3^k} \rfloor \text{ is prime for every } k \geq 1\}$$

be Mills' constant. Saito proved its irrationality in [4]. His later reduction [5, Theorem 1.7] states that if A is algebraic, some $\beta = A^{3^m}$ is a totally real cubic Pisot number whose two small conjugates satisfy

$$|\beta_3| < -\beta_2 \leq \min\{|\beta_3|^{17/23}, \beta^{-17/40}\}. \tag{1}$$

The trace is then the floor at all sufficiently large cubic-power levels. Excluding every such algebraic candidate would prove unconditional transcendence of A. Saito's Problem 1.1 in [3] asks the stronger question: does every cubic Pisot number have infinitely many composite cubic-power floors?

The word "question" matters here. The source papers contain explicit problems and alternatives, and an answer to a modified sequence does not automatically settle the original least constant. All source theorem numbers below refer to the specified arXiv versions, since journal numbering may differ.

1.1 What is already resolved and what remains

Source and item	Mathematical target	Status used in this paper
[4], Q1.6	Irrationality of the least constant with exponent 2^k	Not proved here.
[4], Q1.7	The cubic Pisot power alternative for Mills' constant	Not excluded in full.
[4], Q5.3	RH implies Mills transcendence	Already answered by Saito; his later Theorem 1.8 uses the Density Hypothesis.
[3], P1.1	All cubic Pisot floors at 3^n	Restricted closures and quantitative exceptions below; full problem remains.
[3], P1.7–1.8	Nonreversible clocks and shifted dyadic Fibonacci values	Already treated in the supplied older manuscripts [008] and [007].
[6], Q1.4–1.5	Unbounded multiplier and Wright-set dimensions	Already treated in supplied older [031], Theorems 14.3 and 15.1.

The last two rows report the content and audited proofs of the user-supplied manuscripts, not a claim of journal acceptance or historical priority. They are not counted again as new resolutions. The RH implication is a published external result of Saito, not a consequence newly obtained from the present work.

1.2 Contribution relative to the supplied manuscripts

The new attachment inventory contains 244 TeX manuscripts, including 38 in the Mills–Pisot topic. We focused on their arithmetic proofs and on plausible transfers from the other topics; this is not a claim to have independently re-proved all 244 manuscripts. The core additions and their limitations are as follows.

1. Theorem 2.2 synchronizes finitely many shifts on one progression. The old single-shift positive-density statements alone do not supply this synchronization.

2. Theorem 4.1 extends the unshifted cubic coefficient sieve of round S37 to every shift and every odd degree, uniformly in the shift, with an improved explicit range of constants. The older [031] already contains qualitative density-one results; the quantitative rate and the stated extensions are the additions.

3. Theorem 5.1 adds three exact witnesses to the Pell cover of rounds S35–S36. It removes two further valuation layers, but leaves an infinite suborbit to be analyzed.

4. Theorem 6.4 refines the finite patterns of S28 by specifying exact selected prime valuations and their periodic recurrence. Prime-designated positions still require Schinzel's hypothesis H. The all-index irreducibility of S27 is reproduced as prior input, not counted as new.

5. Theorem 7.1 quantifies the persistence of Saito-balanced units in a fixed congruence coset, explaining why finite local exclusions and the real size conditions can coexist.

6. Theorems 9.3 and 9.5 turn an inherited cubic local obstruction into exclusions for all rational powers of Mills' constant in larger number fields. Theorem 8.1 records the existing cyclic norm restriction in a positive-density formulation for both floor branches.

7. Theorem 10.1 removes the seed assumption from finite-prime survival by allowing a fixed nonzero integral norm. It applies to every remaining totally real cubic field and every prescribed finite prime set. Corollary 10.2 places such survivors in any real interval after normalization.

The manuscript contains mathematical proofs and exact finite certificates; external priority remains unestablished. In particular, none of the remaining numbered global questions in the first four rows is claimed as a new complete resolution.

1.3 Conventions

A composite integer is positive and larger than one, with a proper prime divisor. Finite initial exceptions are harmless only after growth has been proved. Lower density always refers to the index n, not to the enormous integer value of the trace. A Pisot number is a real algebraic integer greater than one with every other conjugate strictly inside the unit circle. Traces count all conjugates with multiplicity when polynomial specializations are used.

2 Simultaneous composite windows from a three-adic limit

For a monic integral cubic $X^3 - uX^2 + vX - N$, let U_n be its 3^n-th root power sum, let V_n be the corresponding second symmetric function, and let $E_n = N^{3^n}$. Newton's identities give

$$(U_{n+1}, V_{n+1}, E_{n+1}) = (U_n^3 - 3U_nV_n + 3E_n,\ V_n^3 - 3E_nU_nV_n + 3E_n^2,\ E_n^3). \tag{2}$$

The norm coordinate must be advanced, except when $N = 0, 1, -1$.

Lemma 2.1 (Three-adic contraction)**.** *The sequence U_n has a limit $\lambda \in \mathbb{Z}_3$, with $\lambda - U_n \in 3^{n+1}\mathbb{Z}_3$. This limit depends only on $(u, v, N) \pmod 3$.*

Proof. Let F denote the polynomial map in (2). It is the identity modulo three. If $\mathbf{x} \equiv \mathbf{y} \pmod{3^r}$, $r \geq 1$, expansion gives $F(\mathbf{x}) \equiv F(\mathbf{y}) \pmod{3^{r+1}}$; every first derivative has coefficients divisible by three, and higher terms have the requisite divisibility. Consecutive iterates therefore differ by a vector divisible by 3^{n+1} at step n. The same argument for two initial points congruent modulo three proves equality of their limits. □

Theorem 2.2 (A common progression for finitely many shifts)**.** *Suppose $U_n \to +\infty$ in the real topology. Let $\mathcal{H} \subset \mathbb{Z}$ be finite and assume*

$$\lambda + h \notin \{1, -1\} \qquad (h \in \mathcal{H}).$$

There are integers $n_0, D \geq 1$ such that all $U_{n_0+jD} + h$, $h \in \mathcal{H}$, are composite for every sufficiently large j.

Proof. Choose a common large n_0 so that $Q_h = U_{n_0} + h > 1$ for all shifts. When $3 \nmid \lambda + h$, also arrange

$$v_3(Q_h - 1) < n_0, \qquad v_3(Q_h + 1) < n_0.$$

The two limiting valuations are finite by hypothesis, so Lemma 2.1 permits this choice. If $3 \mid \lambda + h$, the prime three divides $U_n + h$ at every level and supplies a recurring divisor.

For any other shift factor Q_h into primes. If every prime factor q satisfied $v_3(q^2 - 1) \geq n_0$, then each such q would be 1 or -1 modulo 3^{n_0}. Their product, including multiplicities, would make $Q_h \equiv \pm 1 \pmod{3^{n_0}}$, a contradiction. Hence there is a prime $q_h \mid Q_h$ with

$$v_3(q_h^2 - 1) \leq n_0 - 1.$$

This prime is not three. Every nonzero root of the cubic modulo q_h has order dividing $M_h = \operatorname{lcm}(q_h - 1, q_h^2 - 1, q_h^3 - 1)$. Elementary lifting of valuations gives

$$v_3(M_h) \leq v_3(q_h^2 - 1) + 1 \leq n_0.$$

After powering to 3^{n_0}, all remaining root orders are prime to three. A period D_h for cubing on these groups therefore gives $q_h \mid U_{n_0+jD_h} + h$ for every $j \geq 0$. Zero roots and repeated roots cause no difficulty; the power sum counts multiplicities.

Take D to be the least common multiple of the finitely many D_h, with $D = 1$ when there are none. All chosen prime divisors recur on the same progression. Real growth makes them proper divisors for all sufficiently large j. Unlike a prime-seed argument, this construction works even when every initial value selected is already composite. □

Corollary 2.3 (Windows around cubic Pisot floors)**.** *Let β be a cubic Pisot number and let λ be its three-adic trace limit. If λ is not an ordinary integer, then for every $H \geq 0$ there is an arithmetic progression of indices on which, eventually, all*

$$\lfloor \beta^{3^n} \rfloor + h, \qquad |h| \leq H,$$

are composite simultaneously. Nonreal conjugates are permitted.

Proof. Apply Theorem 2.2 to all integer shifts in $[-H-1, H]$. None can give $\lambda + h = \pm 1$. The sum of the two small conjugate powers tends to zero, so the floor is eventually either U_n or $U_n - 1$. The stated floor window lies inside the chosen trace window, regardless of changes in the sign of the small-conjugate sum. □

Separate positive-density statements for different shifts cannot be intersected without an argument. The common initial index and the prime-divisor construction above provide that missing step. The single-shift order-budget mechanism was already present in the older manuscript [15]; the simultaneous progression is the strengthening here.

Proposition 2.4 (Exact residue-class scope)**.** *The limit λ is an ordinary integer precisely when the residual cubic over $\mathbb{F}_3$ either splits completely or is $(X - r)(X^2 + 1)$ with $r \in \mathbb{F}_3$. Thus Corollary 2.3 applies to fourteen of the twenty-seven coefficient triples modulo three.*

Proof. For each nonzero residual root, take its Teichmüller lift, the unique prime-to-three root of unity with that residue. In a three-adic splitting field an integral root of positive valuation vanishes under repeated cubing; a unit root's principal-unit part tends to one. Frobenius permutes the multiset of residual roots, so λ is the sum of the Teichmüller lifts, with multiplicity.

A linear root contributes $0, 1$, or -1. The irreducible quadratic $X^2 + 1$ contributes two fourth roots of unity with sum zero. The other two irreducible quadratics are $X^2 + X - 1$ and $X^2 - X - 1$. Their lifted pairs have sum with square -2, so adding one linear contribution cannot give an integer.

For an irreducible cubic, the nonzero roots have order thirteen or twenty-six. The lifted trace is $\pm(\zeta^a + \zeta^{3a} + \zeta^{9a})$, where ζ is a primitive thirteenth root and $a \neq 0 \pmod{13}$. This is not rational: otherwise a polynomial of degree at most twelve with at most four nonzero coefficients would vanish at ζ, in contradiction to its minimal polynomial $1 + X + \cdots + X^{12}$. There are ten completely split monic cubics (multisets of three roots from a three-element set), three of the second integer-limit type, six with the other quadratic factors, and eight irreducible cubics. These exhaust the twenty-seven cases. □

Integer limit λ	$(u, v, N) \pmod 3$
-3	$(0,0,2)$
-2	$(1,1,0)$
-1	$(2,0,0),\ (2,1,2),\ (2,2,1)$
0	$(0,0,0),\ (0,1,0),\ (0,2,0)$
1	$(1,0,0),\ (1,1,1),\ (1,2,2)$
2	$(2,1,0)$
3	$(0,0,1)$

The complement of this table is the fourteen-class scope. This is a count of residue classes, not a density assertion about Pisot numbers under an unspecified ordering. For example, the plastic number, the real root of $X^3 - X - 1$, has irreducible reduction modulo three and satisfies the simultaneous-window conclusion.

For the unshifted trace problem only the six classes with $\lambda = \pm 1$ can escape Theorem 2.2. These include the surviving Mills alternatives. For the predecessor trace $U_n - 1$, only the four classes with $\lambda = 0$ or 2 can escape; all four have $N \equiv 0 \pmod 3$. In particular, whenever $3 \nmid N$ and the floor is eventually $U_n - 1$, its composite indices have positive lower density. This last single-shift consequence was already available in the earlier local obstruction.

3 Field criteria and a uniform density bound

The preceding simultaneous-return theorem has a useful field-theoretic consequence. We first record an elementary bound that also handles changing signs of nonreal conjugates.

Proposition 3.1 (A parity and trace criterion)**.** *Let β be a cubic Pisot number of odd norm. If $\mathrm{Tr}(\beta) \equiv 0 \pmod 3$, then*

$$\liminf_{M\to\infty} \frac{1}{M} \#\{0 \le n < M : \lfloor \beta^{3^n} \rfloor \text{ is composite}\} \ \ge \ \frac{1}{2}.$$

The two small conjugates may be nonreal.

Proof. Let U_n, V_n, E_n be the coefficient sequences in (2). Since the norm is odd, $E_n = 1$ modulo two. The map on $(U_n, V_n) \pmod 2$ is

$$(a, b) \longmapsto (a + ab + 1,\ b + ab + 1).$$

Its four states satisfy

$$00 \longmapsto 11, \qquad 11 \longmapsto 11, \qquad 01 \longleftrightarrow 10.$$

Consequently U_n is odd either at every sufficiently large index or at exactly one parity of the indices. At least half of all indices therefore have U_n odd. At the same time, (2) modulo three gives $U_n \equiv U_0 \equiv 0 \pmod 3$ for every n.

The sum of the two small conjugate powers is real and tends to zero. Thus, for every sufficiently large n, $\lfloor \beta^{3^n} \rfloor$ belongs to $\{U_n, U_n - 1\}$. When U_n is odd, the first possibility is divisible by three and the second is divisible by two. Growth makes these divisors proper. This proves the density bound without any assumption on the sign pattern of the small-conjugate sum. □

Theorem 3.2 (Cubic fields without a three-adic embedding)**.** *Let K be a cubic number field admitting no field embedding into $\mathbb{Q}_3$. Let $\beta \in \mathcal{O}_K$ be a cubic Pisot number with $3 \nmid N_{K/\mathbb{Q}}(\beta)$. Then the composite indices of $\lfloor \beta^{3^n} \rfloor$ contain, up to finitely many terms, an arithmetic progression. In particular they have positive lower density.*

More precisely, write f for the minimal polynomial of β. If $f \bmod 3$ is irreducible, every finite window about these floors is simultaneously composite on a progression, as in Corollary 2.3. Otherwise $f \bmod 3$ is a cube of a nonzero linear factor. In this second case, if the norm is also odd, the lower density is at least $1/2$.

Proof. A simple root of $f \bmod 3$ would lift by Hensel's lemma to a root in $\mathbb{Q}_3$. Since $K = \mathbb{Q}(\beta)$, that root would define an embedding $K \hookrightarrow \mathbb{Q}_3$, contrary to hypothesis. A monic cubic over $\mathbb{F}_3$ without a simple linear factor is either irreducible or $(X - a)^3$. The norm assumption makes $a \neq 0$ in the latter case.

In the irreducible case Proposition 2.4 gives a noninteger three-adic trace limit, and Corollary 2.3 proves the asserted window statement. In the remaining case the three Teichmüller lifts

of the residual roots are all 1 or all -1. Thus the trace limit is $\lambda = 3$ or $\lambda = -3$, respectively. Neither λ nor $\lambda - 1$ equals 1 or -1. Apply Theorem 2.2 to the two shifts $\{0, -1\}$. Both U_n and $U_n - 1$ are composite on one eventual progression; the Pisot estimate again places the floor among these two integers. Finally $(X - a)^3$ has zero quadratic coefficient modulo three, so the odd-norm case satisfies Proposition 3.1. □

Corollary 3.3 (Totally ramified cubic fields)**.** *If three is totally ramified in a cubic number field K, then every cubic Pisot unit of K has composite floors at a set of indices of lower density at least* 1/2*.*

Proof. The algebra $K \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is a field of degree three, so K has no embedding into $\mathbb{Q}_3$. For any $\beta \in \mathcal{O}_K$, its three local conjugates have the same residue in the residue field $\mathbb{F}_3$. Its minimal polynomial therefore reduces to $(X - a)^3$ modulo three. A unit has nonzero residue and norm ± 1. The last assertion of Theorem 3.2 applies. □

For example, $K = \mathbb{Q}(r)$ with $r^3 - 3r - 1 = 0$ is a totally real cyclic cubic field: the polynomial is irreducible, has discriminant 81, and its translate $X^3 + 3X^2 - 3$ is Eisenstein at three. Hence the corollary covers every Pisot unit in this field. It also covers every Pisot unit in $\mathbb{Q}(\sqrt[3]{2})$, since $(X - 1)^3 - 2 = X^3 - 3X^2 + 3X - 3$ is Eisenstein at three. An explicit example in the latter field is $\beta = 1 + \sqrt[3]{2} + \sqrt[3]{4}$, whose minimal polynomial is $X^3 - 3X^2 - 3X - 1$. It has norm one and a unique real root greater than one; its conjugate pair has modulus $\beta^{-1/2} < 1$. Thus the density bound applies even though the floor correction need not have an eventually constant sign.

Completion of the norm-positive case in the field of discriminant 1524**.** Let r be the largest root of $X^3 - 7X^2 - X + 1$ and let $K = \mathbb{Q}(r)$. Showa round 20 proves that every norm-one unit $\beta \in K$ with $\mathrm{Tr}(\beta^{3^n}) \to +\infty$ has a positive lower density of composite trace terms. This is a prior result of the supplied archive, not a new trace theorem here. It also yields the following precise floor statement.

Corollary 3.4. *Every cubic Pisot unit of norm one in the field of discriminant* 1524 *has a positive lower density of composite floors $\lfloor \beta^{3^n} \rfloor$.*

Proof. The field is totally real. Since $\beta > 1$ and its norm is one, its two small conjugates have the same sign. If both are negative, the floor equals the trace eventually, and the cited round-20 theorem applies. If both are positive, the floor equals $U_n - 1$ eventually. The four-state parity calculation in the proof of Proposition 3.1 shows that U_n is odd at a set of indices of density at least 1/2. At those sufficiently large indices $U_n - 1$ is a positive even composite integer. □

Scope and comparison with the supplied manuscripts. The old manuscript [15] already proves infinitely many composite floors for every nonreal cubic Pisot number, and gives positive-density results for fixed real shifts. Its projective return theorem already includes the cubic-residue norm refinement; that refinement is not new here. The absence of a three-adic embedding was already a necessary obstruction for a hypothetical prime trace tail in [16]. The new use here is the simultaneous treatment of the two possible floor corrections, together with the explicit density-1/2 criterion. In particular the statements above do not depend on an unjustified intersection of separate positive-density sets.

The discriminant-1524 corollary is a completion from traces to all norm-one Pisot floors using a prior field-specific theorem. It makes no assertion about norm-minus-one units in that field: round 17 exhibits such units that survive several local tests. Nor do these criteria exclude every totally real cubic Pisot candidate in Saito's Problem 1.1 or establish Mills transcendence.

4 Shifted traces: exact local counts and sparse exceptions

This section extends the unshifted cubic sieve of submitted round S37. The extensions are uniformity in the integer shift, both floor branches in totally real cubic fields, and arbitrary odd degree. We include the proof, since changing a trace to a shifted trace is not automatically legitimate in a fixed-norm argument.

For an odd integer $d \geq 3$, write

$$f_{\mathbf{a}}(X) = X^d - a_1 X^{d-1} + a_2 X^{d-2} - \cdots - a_d, \qquad P_n(\mathbf{a}) = \sum_{j=1}^{d} \alpha_j^{3^n}.$$

Here the roots are counted with multiplicity, and Newton's identities define $P_n \in \mathbb{Z}[a_1, \ldots, a_d]$ even at reducible specializations. Its total degree is 3^n, and its unique term of that degree is $a_1^{3^n}$. For a sequence of integers, a *recurring-divisor certificate* is a prime dividing all terms on an infinite arithmetic progression of indices. If the terms tend to positive infinity, such a certificate gives positive lower density of composite terms.

Theorem 4.1 (Uniform shifted sieve)**.** *Put $c_0 = 1/(e\sqrt{2\log 3})$. Fix an odd $d \geq 3$ and $0 < c < c_0$. Let $\mathcal{B} \subset \mathbb{Z}^d$ be any box with side lengths $X_1, \ldots, X_d \geq 3$, and put $m = \min X_i$ and $V = \prod X_i$. For every integer h, all but*

$$O_{d,c}(V \exp(-c\sqrt{\log m})) \tag{3}$$

coefficient vectors in $\mathcal{B}$ have a recurring-divisor certificate for $P_n(\mathbf{a}) + h$. The constant is independent of h and of the location of the box.

There is a constant $C_{d,c} > 0$ such that the certificate can be chosen with prime $p \leq \exp(C_{d,c}\sqrt{\log m})$ and progression period at most $\exp(dC_{d,c}\sqrt{\log m})$. Consequently, whenever $P_n(\mathbf{a}) \to +\infty$, the lower density of composite shifted terms is positive, and is at least the reciprocal of that period.

The theorem gives an upper bound on a set of possible exceptions; it does not prove that this set is empty. The assertion is about the full coefficient space. A thin fixed field, fixed norm, or selected one-parameter family requires a separate argument.

Lemma 4.2 (Exact shifted local count)**.** *For every prime $p \equiv 2 \pmod 3$, $n \geq 1$, and $t \in \mathbb{F}_p$,*

$$\#\{\mathbf{a} \in \mathbb{F}_p^d : P_n(\mathbf{a}) = t\} = p^{d-1}.$$

Repeated roots and zero constant terms are allowed.

Proof. The coefficient vectors are Frobenius-stable orbits of ordered root tuples under S_d. Twisted orbit counting gives the number in question as $d!^{-1} \sum_{\sigma \in S_d} \#\operatorname{Fix}(F\sigma)$, where the fixed tuples also satisfy the power-sum condition. This is an identity of integer counts, including orbits with stabilizers. All tuples under consideration lie in a common finite extension, so ordinary finite-set counting applies.

A cycle of length j in σ corresponds to an element $b \in \mathbb{F}_{p^j}$ and contributes $\operatorname{Tr}_{\mathbb{F}_{p^j}/\mathbb{F}_p}(b^{3^n})$ to the power sum. Every permutation of an odd number of letters has an odd cycle. For such a cycle j, $3 \nmid p^j - 1$, so $b \mapsto b^{3^n}$ is a bijection of $\mathbb{F}_{p^j}$. The field trace is surjective, with fibers of size p^{j-1}. Fixing all other cycles leaves exactly that many choices for this one, giving p^{d-1} fixed tuples for each permutation. Averaging proves the claim. Surjectivity of field trace does not require $p \nmid j$. □

Lemma 4.3 (Geometric separation)**.** *Over an algebraically closed field of characteristic different from three, $P_n - t$ is irreducible for every $n \geq 1$ and every t. For distinct indices i, j and any $s, t \in \mathbb{F}_p$, $p \neq 3$,*

$$\#\{P_i = s,\ P_j = t\} \leq 3^{i+j} p^{d-2}.$$

Proof. Under the finite, surjective Vieta map the first polynomial pulls back to $Z_1^r + \cdots + Z_d^r - t$, $r = 3^n$. Regard this as a polynomial in Z_d over $k[Z_1, \ldots, Z_{d-1}]$. The nonconstant polynomial $Z_1^r + \cdots + Z_{d-1}^r - t$ has a prime factor with multiplicity one: it is squarefree, since any repeated factor would divide all its partial derivatives, which are multiples of the relatively prime monomials $Z_1^{r-1}, \ldots, Z_{d-1}^{r-1}$. Eisenstein at such a factor proves irreducibility of the pullback, and hence of $P_n - t$.

Different indices have different degrees, so the two hypersurfaces have no common component. Their intersection has pure dimension $d-2$ and degree at most 3^{i+j}, by Bézout. An affine variety of dimension r and degree D has at most Dp^r rational points. One proof is induction on dimension: on each irreducible component choose a nonconstant coordinate, slice at its p rational values, and apply the degree bound for hyperplane sections. Constant coordinates are discarded. This argument also handles components defined only over an algebraic closure, since it bounds their $\mathbb{F}_p$-points individually. These facts give the displayed bound. □

Lemma 4.4 (A uniform return budget)**.** *Fix $B \geq 1$ and put $b_d = \lfloor \log_3(d/2) \rfloor$. Suppose $p \equiv 2 \pmod 3$ and $v_3(p+1) \leq B$. For every coefficient vector modulo p, its power-sum sequence is periodic from index $B + b_d$, with period less than p^d.*

Proof. For an irreducible root factor of degree j, each nonzero root has order dividing $p^j - 1$. For odd j this order is prime to three; for even $j = 2k$, the lifting-the-exponent identity gives

$$v_3(p^{2k} - 1) = v_3(p+1) + v_3(k) \leq B + b_d.$$

After that many cubings all nonzero root powers have orders prime to three. Cubing is then a permutation of their finite groups. Zero roots stay zero and multiplicities do not change this argument. The least common multiple M of the remaining root orders is at most the product of $p^j - 1$ over distinct nonzero irreducible factors, which is less than p^d. A common period is the order of 3 modulo M, with period one when $M = 1$. □

Proof of Theorem 4.1. The eligible primes of Lemma 4.4 form fixed reduced residue classes modulo 3^{B+1} of density

$$\delta = \delta_B = \frac{1 - 3^{-B}}{2}.$$

Let $k_0 = B + b_d$ and use the L consecutive levels $k_0, \ldots, k_0 + L - 1$. At an eligible prime let Ω_p be the union of the loci $P_n + h = 0$ at these levels, and put $\omega(p) = |\Omega_p|$. Exact counts and the first two inclusion–exclusion inequalities give

$$\frac{L}{p} - \frac{D_L}{p^2} \leq \frac{\omega(p)}{p^d} \leq \frac{L}{p}, \qquad D_L = \sum_{k_0 \leq i < j < k_0 + L} 3^{i+j} < 9^{k_0+L}.$$

For $p > y := 9^{k_0+L+1}$ we therefore have

$$\frac{L-1}{p} \leq \frac{\omega(p)}{p^d} \leq \frac{L}{p}. \tag{4}$$

These estimates are independent of the shift, even when it varies with the box.

We use the classical multidimensional large sieve in the form

$$|S| \ll_d \frac{(m + Q^2)^d}{H(Q)}, \qquad H(Q) = \sum_{\substack{q \leq Q \\ q \text{ squarefree}}} \prod_{p \mid q} \frac{\omega(p)}{p^d - \omega(p)}, \tag{5}$$

for a set S in a cube of side m avoiding the forbidden classes. For completeness, its normalization can be seen directly. On $(\mathbb{Z}/p\mathbb{Z})^d$ take a mean-zero function equal to 1 on the allowed classes

and to $-(p^d-\omega(p))/\omega(p)$ on the forbidden classes. Its normalized squared norm is $1/g(p)$, where $g(p)=\omega(p)/(p^d-\omega(p))$. The Chinese-remainder product, Parseval, and Cauchy–Schwarz imply

$$g(q)|S|^2 \le \sum_{\substack{\mathbf{a} \bmod q \\ \gcd(a_1,\dots,a_d,q)=1}} \left|\sum_{\mathbf{x}\in S} e^{2\pi i \mathbf{a}\cdot\mathbf{x}/q}\right|^2.$$

Primitive frequencies with denominators at most Q are Q^{-2}-separated. The analytic large sieve bounds their combined energy by $O_d((m+Q^2)^d)|S|$, proving (5). This is a classical input, not a new sieve inequality; see [7].

Set $Q=\lfloor\sqrt{m}\rfloor$, $s=\sqrt{\log Q}$, $L=\lfloor as\rfloor$, and $z=\exp(bs)$, with fixed positive a,b such that

$$b > a\log 9, \qquad ab\delta < 1.$$

Only eligible primes in $(y,z]$ are used. The prime number theorem in these fixed residue classes and partial summation give

$$\sum_{y<p\le z}^{*}\frac{1}{p} = \delta\log\frac{b}{a\log 9}+o(1), \qquad \sum_{p\le z}^{*}\frac{\log p}{p} = (\delta+o(1))\log z.$$

The star restricts to eligible primes. The moving endpoints cause no uniform-modulus issue: B is fixed first and both endpoints tend to infinity.

Expand the finite Euler product $W=\prod^{*}_{y<p\le z}(1+g(p))$ with its nonnegative squarefree weights. Equation (4) implies

$$\log W \ge \left(a\delta\log\frac{b}{a\log 9}+o(1)\right)s.$$

The weighted mean of $\log q$ in this expansion is exactly

$$\sum_{y<p\le z}^{*}\frac{g(p)}{1+g(p)}\log p \le (ab\delta+o(1))\log Q.$$

Since $ab\delta<1$, Markov's inequality leaves a fixed positive fraction of the weight on $q\le Q$. Thus $H(Q)\gg W$. Take

$$a=\frac{1}{e\sqrt{\delta\log 9}}, \qquad b=(1-\varepsilon)e\sqrt{\frac{\log 9}{\delta}}$$

with $\varepsilon>0$ sufficiently small. As $\varepsilon\downarrow 0$, the coefficient in the lower bound for $\log H(Q)$, when expressed in units of $\sqrt{\log m}$, approaches $e^{-1}\sqrt{\delta/\log 3}$. Finally choose a fixed large B so that this exceeds the prescribed $c<c_0$. We obtain $H(Q)\gg_{d,c}\exp(c\sqrt{\log m})$.

The cube sieve proves (3) for a cube. A rectangular box is covered by at most $2^dV/m^d$ such cubes. A point not in the sifted set has a zero at one of the chosen levels modulo some $p\le z$. Lemma 4.4 supplies the claimed certificate and period bound. Bounded values of m are absorbed in the constants. □

Corollary 4.5 (Both totally real floor branches)**.** *Among coefficient triples in any box of minimum side m, the totally real cubic Pisot polynomials for which $\lfloor\beta^{3^n}\rfloor$ fails to be composite on a set of positive lower density number at most*

$$O_c(X_1X_2X_3e^{-c\sqrt{\log m}}), \qquad c<c_0.$$

Proof. If the two small real conjugates are γ,η, their odd-power sum has a constant nonzero sign for all sufficiently large powers. The only potential exception is $\gamma=-\eta$, which would make β the integer trace, contradicting cubic irreducibility. Thus the floor is eventually P_n or P_n-1. Apply the theorem to both shifts and take a union bound. The traces tend to infinity. □

Corollary 4.6 (Many shifts)**.** *For any finite set $\mathcal{H} \subset \mathbb{Z}$, the exceptional count for having a certificate for every $h \in \mathcal{H}$ is at most $O_{d,c}(|\mathcal{H}|Ve^{-c\sqrt{\log m}})$. In particular, for $0 < \eta < c < c_0$, one may take $|h| \le \exp(\eta\sqrt{\log m})$ and retain a saving $e^{-(c-\eta)\sqrt{\log m}}$. The progressions for different shifts in this corollary need not coincide. A genuinely simultaneous progression is proved separately in Section 2.*

5 Two further valuation layers on the twice-square Pell locus

Consider

$$g_u(X) = X^3 - uX^2 - X + 1, \qquad 4u^2 + 5u + 23 = 2w^2, \quad u \ge 2, \quad w > 0.$$

Let $P_n(u)$ be the sum of the 3^n-th powers of the roots of g_u. Signs at $-1, 0, 1, u, u+1$ give roots $-x, y, \beta$ with $0 < x, y < 1 < u < \beta < u + 1$. Neither rational candidate ± 1 is a root. The norm is -1, and the second symmetric function -1 gives $\beta(y - x) = -1 + 1/\beta < 0$. Thus $x > y$ and $P_n(u) = \lfloor \beta^{3^n} \rfloor$ at every level. Write

$$T(u, w) = (17u + 12w + 10, 24u + 17w + 15), \qquad (a_j, r_j) = T^j(9, 14), \quad (b_j, s_j) = T^j(1, 4).$$

The Pell reduction in the supplied round 35 identifies the entire locus with a_j for $j \ge 0$ and b_j for $j \ge 1$. Round 36 excludes the first orbit and all second-orbit indices except $j \equiv 0, 4374 \pmod{13122}$. We have independently replayed the complete finite certificate for that assertion. The following three additional witnesses strengthen its conclusion.

Theorem 5.1. *If $(P_n(u))_{n\ge 0}$ is eventually prime on this Pell locus, then*

$$u = b_j, \qquad 39366 \mid j, \qquad v_3(u - 1) \ge 10, \qquad u > 10^{59049}.$$

Every parameter excluded by this condition has a positive lower density of composite trace indices.

Proof. Put $\eta = 17 + 12\sqrt{2}$ and write $j = 2k$. The identity

$$(8b_j + 5) + 4s_j\sqrt{2} = (13 + 16\sqrt{2})\eta^j$$

shows that $\eta^{2m} = 1 \pmod p$ makes b_{2k} periodic modulo p with period dividing m. The following table consists of exact certificates; its column c is $b_{2a} \bmod p$.

p	m	a	c	n	$v_3((p-1)(p^2-1)(p^3-1))$
8279983	6561	2187	7051991	4863973	25
27398737	19683	6561	15563564	4929815	31
15037811	19683	13122	1777082	549606	9

In every row, p is prime, $\eta^{2m} = 1 \pmod p$, and $P_n(c) = 0 \pmod p$. The root-order return lemma makes each such zero recurrent, because n exceeds the last column. Thus the entire parameter-index class $k \equiv a \pmod m$ has infinitely many composite traces, along an arithmetic progression of trace indices.

For reproducibility, put $M_p = (p-1)(p^2-1)(p^3-1)$ and $E = 3^n \bmod M_p$. Binary polynomial powering modulo (g_c, p) gives $X^E = A + BX + CX^2$, with coefficient triples in the order of the table

$$(7459560, 4961071, 5552814), \qquad (21974350, 24684017, 2338784),$$
$$(13021067, 3973052, 8155287).$$

Each satisfies $3A + cB + (c^2 + 2)C = 0 \pmod p$. This verifies a large trace-index witness without constructing the integer trace. Primality of the three modest witness primes is certified by trial division up to their square roots.

The round 36 cover leaves $k \equiv 0, 2187 \pmod{6561}$. The first new row removes the latter class. The two remaining rows remove $k \equiv 6561, 13122 \pmod{19683}$, leaving $19683 \mid k$. Therefore $39366 \mid j$. The exact identity $v_3(b_j - 1) = 1 + v_3(j)$ for positive even j now gives $v_3(u-1) \geq 10$. On the Pell curve, $w > 4u/3$, so $T(u,w)_1 > 33u$. Hence

$$u = b_j > 33^j \geq 33^{39366} = 1089^{19683} > 10^{59049}.$$

□

Corollary 5.2 (Relative count on the complete Pell locus)**.** *Let $\mathcal{P}(X)$ count the parameters in the displayed Pell locus with $u \leq X$, and let $\mathcal{E}(X)$ count those with an eventually prime trace tower. Then*

$$\mathcal{P}(X) = \frac{2\log X}{\log(17 + 12\sqrt{2})} + O(1), \qquad \mathcal{E}(X) \leq \frac{\log X}{39366 \log(17 + 12\sqrt{2})} + O(1).$$

In particular $\limsup_{X\to\infty} \mathcal{E}(X)/\mathcal{P}(X) \leq 1/78732$.

Proof. Each affine Pell orbit grows as a positive constant times $(17 + 12\sqrt{2})^j$. The theorem retains at most every 39366-th index of the second orbit and none of the first. □

Remark 5.3 (The remaining uniform step)*.* The strengthened numerical and valuation bounds concern this special parameter family, not the numerical value of Mills' constant. The exceptional suborbit is not proved empty. Indeed, for any fixed modulus M, let L be the period of the orbit of $(1,4)$ under T modulo M. Such a period exists because the linear part of T has determinant one. At every positive index divisible by $39366L$, $b_j \equiv 1 \pmod{M}$. The trace pair $(1,-1)$ is an exact fixed point of the cubing recurrence, so $P_n(b_j) \equiv 1 \pmod{M}$ for every n. Thus an arbitrary prescribed finite divisibility sieve still leaves unboundedly many parameters in the surviving branch.

6 Uniform irreducibility and exact finite divisibility patterns

Provenance and improvement. The supplied Showa round 27 already proves uniform irreducibility in an explicit cyclic cubic unit family. It supersedes the earlier six-term conditional prefix result as an existence theorem, although the earlier six polynomials belonged to a different monomial family. Round 28 already realizes every finite prime/composite pattern under Schinzel's hypothesis H. We reproduce the decisive proofs and add two refinements: a general signed-specialization criterion with an exact three-block limitation, and arbitrary prescribed local multiplicities which recur unchanged on infinite progressions of trace indices. These refinements do not remove hypothesis H from any assertion of simultaneous prime values.

6.1 The balanced unit and its all-index irreducibility

Put

$$A = a^3 + 2a^2 + 3a + 3, \qquad Q = a^2 + a + 1,$$
$$f_a(Z) = Z^3 - AZ^2 - a^2 Z + 1, \qquad d(a) = (a+1)(a^2+3)(a^2+3a+3).$$

The discriminant is $d(a)^2$. In the cyclic root ordering set

$$s = \frac{ar+1}{1-Qr}, \qquad t = \frac{r-1}{Qr+a}, \qquad \beta_a = -r^5 s.$$

For integer $a \geq 8$, take r to be the largest root. The rational-root test and square discriminant show that the field is cyclic cubic. The trace seed is

$$\begin{aligned} U_0 &= a^{14} + 9a^{13} + 45a^{12} + 161a^{11} + 450a^{10} + 1020a^9 + 1918a^8 \\ &\quad + 3021a^7 + 3987a^6 + 4379a^5 + 3942a^4 + 2820a^3 + 1521a^2 + 558a + 105, \\ V_0 &= -3a^6 - 15a^5 - 40a^4 - 75a^3 - 105a^2 - 90a - 33, \end{aligned}$$

and

$$U_{n+1} = U_n^3 - 3U_nV_n - 3, \qquad V_{n+1} = V_n^3 + 3U_nV_n + 3. \tag{6}$$

These identities follow from $rst = -1$ and Newton's identities: $U_0 = -T_{5,1}$ and $V_0 = -T_{4,5}$, where $T_{ij} = r^is^j + s^it^j + t^ir^j$. Thus $U_n(a) = \mathrm{Tr}(\beta_a^{3^n})$.

Theorem 6.1 (Showa round 27, independently audited)**.** *Every U_n is monic and irreducible over $\mathbb{Q}$, of degree $14 \cdot 3^n$, and $U_n(-1) = 1$. For every integer $a \geq 8$ the number β_a is a cubic Pisot unit of norm -1 whose conjugates are $\beta_a, -X, Y$, with*

$$0 < Y < X < 1, \quad X^{40}\beta_a^{17} < 1, \quad X^{23} < Y^{17}, \qquad U_n(a) = \lfloor \beta_a^{3^n} \rfloor.$$

Proof. First every odd trace is positive at every real parameter. For real $a \neq -1$, the three roots of f_a are distinct and real. Their product is -1 and their pairwise sum is $-a^2 \leq 0$, so they have the form $p > q > 0, -w < 0$ with $pqw = 1$. If $a > -1$, then $f_a(1) < 0$ gives $q < 1 < p$. The cyclic fractional-linear map sends p to $-w$: its value cannot vanish or have a pole on this connected parameter interval, since its determinant is $(a+1)^2 \neq 0$, and it is negative at $a = 0$. The conjugates of β_a are therefore

$$P = p^5w, \qquad R = w^5q, \qquad -B = -q^5p, \qquad B/P = q^6/p^3 < 1.$$

Their odd-power sum is positive. If $a < -1$, then $f_a(1) > 0$ and $A(a) < A(-1) = 1$ show that both positive roots lie below one: if both exceeded one, $A = p + q - 1/(pq) > 1$. In either cyclic orientation the negative conjugate has magnitude $B < 1$, while $PRB = 1$ forces a positive conjugate above one. The odd trace is again positive. At $a = -1$ the specialized conjugate multiset is $(1, 1, -1)$, and every odd trace is one.

Direct reduction gives

$$\overline{U}_0 = a^5(a-1)^3(a^2+1)^3 \quad \text{over } \mathbb{F}_3. \tag{7}$$

For $n \geq 1$ the three Hensel blocks of U_n are irreducible over $\mathbb{Q}_3$ and have degrees $5 \cdot 3^n, 3 \cdot 3^n, 6 \cdot 3^n$. Here is the valuation argument, including repeated residual factors. At a root ξ of U_n, monicity gives integrality and

$$v_3(U_j(\xi)) = 3^{j-n} \quad (0 \leq j < n).$$

The last step follows from $U_{n-1}^3 = 3(1 + U_{n-1}V_{n-1})$; descending steps compare valuations less than one with terms divisible by three. If a residual irreducible factor h has multiplicity m and degree e, write $U_0 = H^mB + 3C$ with $B(\xi)$ a unit. Then $v_3(H(\xi)) = 1/(m3^n)$. The ramification index is divisible by $m3^n$ and the residue degree by e. Their product equals the degree of the Hensel block, proving its irreducibility.

A monic rational factor F of U_n is integral and has no real root, so it has even degree and is positive on the real line. The only proper even subset sums of $5, 3, 6$ are 6 and 8. The corresponding residual factors are

$$(a^2+1)^{3^{n+1}}, \qquad a^{5 \cdot 3^n}(a-1)^{3^{n+1}}.$$

Both have value -1 modulo three at $a = -1$. On the other hand F and its monic complement have positive integer values there multiplying to $U_n(-1) = 1$, so $F(-1) = 1$. This contradiction proves irreducibility for every $n \geq 1$.

For the seed, the exact factorizations modulo two have degrees $6, 8$, and those modulo thirteen have degrees $5, 9$. For reproducibility, the factors are specified by ascending coefficient arrays:

$$\begin{array}{r|l} 2 & (1,0,1,1,0,1,1), \quad (1,0,0,1,0,1,1,0,1) \\ 13 & (12,7,8,8,4,1), \quad (12,7,2,11,12,9,6,4,5,1). \end{array}$$

Their products and irreducibility are certified by the Frobenius criterion $X^{p^d} = X \bmod F$ and $\gcd(F, X^{p^{d/q}} - X) = 1$ for primes $q \mid d$. The disjoint proper factor-degree sets prove rational irreducibility of U_0.

Finally, for $a \geq 8$ and $w = -s$,

$$a^3 < r < 2a^3, \quad a/Q < w < 1/a, \quad 0 < t < 1/a^2, \quad rwt = 1,$$

so

$$(4/5)a^{14} < \beta_a < 32a^{14}, \quad X = t^5 r < 2a^{-7}, \quad Y = w^5 t < a^{-7}.$$

Since $V_0(a) < -1$ and $\beta_a XY = 1$, $\beta_a(Y - X) = V_0(a) + 1/\beta_a < 0$, giving $X > Y$. Also $X^{40}\beta_a^{17} < 2^{125}a^{-42} \leq 1/2$, and $X^{23}/Y^{17} = X^{40}\beta_a^{17}$. Every odd-power trace is its positive dominant power minus a number strictly between zero and one, which proves the floor identities. □

6.2 A signed-specialization criterion and its exact limitation

Proposition 6.2. *Suppose integer polynomials U_0, V_0 satisfy (6), U_0 is monic, $\deg V_0 < 2 \deg U_0$, and for some integer a_* one has $(U_0(a_*), V_0(a_*)) = (1, -1)$. Suppose each U_n is strictly positive on $\mathbb{R}$. Write $\overline{U}_0 = \prod_{i=1}^{r} h_i^{m_i}$ into distinct monic irreducible factors over $\mathbb{F}_3$, with $\deg h_i = d_i$. Give block i the label*

$$\lambda_i = (m_i d_i \bmod 2,\ h_i(a_*)^{m_i}) \in (\mathbb{Z}/2\mathbb{Z}) \times \mathbb{F}_3^{\times}.$$

If no nonempty proper subcollection of these labels has product $(0, 1)$, then U_n is irreducible over $\mathbb{Q}$ for every $n \geq 1$. Moreover, this label criterion is possible only for $r \leq 3$. With $r = 3$, it holds exactly when the labels are the three different nonidentity elements of this group.

Proof. The root-valuation proof above gives one irreducible local block of degree $m_i d_i 3^n$ and reduction $h_i^{m_i 3^n}$ for each i. Since 3^n is odd, its parity and value at a_* retain precisely the label λ_i. A proper rational factor must have even degree and positive integer value one at a_*, contradicting the label condition. The product of all labels is $(0, 1)$, because U_0 has even degree and value one at a_*. The group has three nonidentity elements, each of order two. A minimal product-one sequence cannot contain the identity (unless it has length one), or a repeated element (unless it has length two). Thus it has length at most three; the three-element case must consist of all three nonidentity elements. Conversely those three labels have product one and no proper product-one subcollection. □

This is a limitation of this particular factor obstruction, not a bound on the number of local factors of a globally irreducible polynomial. For the balanced seed the labels are $(1, -1), (1, +1), (0, -1)$, so the criterion is extremal. The four-block degree-31 seed of the earlier six-prefix argument cannot be handled by this exact criterion alone.

6.3 Simple forcing roots and exact local multiplicities

We next state a strengthening of round 28. All prime-value assertions below explicitly assume the usual integer-prime hypothesis H.

Lemma 6.3 (Forcing roots, from round 28)**.** *For every finite window $I = \{0, \dots, L\}$ and $j \in I$, a positive- Dirichlet-density set of primes $p \equiv 2 \pmod 3$ has a residue b such that*

$$U_j(b) = 0, \quad U_j'(b) \neq 0, \quad d(b) \neq 0, \qquad U_i(b) \neq 0 \quad (i \in I \setminus \{j\}).$$

Any prescribed finite set of primes may be excluded.

Proof. Let $K_j = \mathbb{Q}(\xi)$ for a root of U_j. It has an odd-degree completion above three. For $j \geq 1$ this follows from the local blocks of degrees $5 \cdot 3^j$ and $3 \cdot 3^j$. For $j = 0$, Hensel lifts the residual factor a^5 to a degree-five block; at least one of its irreducible constituents has odd degree. Thus $\mathbb{Q}(\sqrt{-3})$ is not a subfield of K_j, since its completion at three has degree two and would embed in every completion of K_j.

In a Galois extension containing the splitting field of U_j and $\mathbb{Q}(\sqrt{-3})$, choose an automorphism fixing ξ and acting nontrivially on the quadratic field. It exists by the preceding noncontainment. Its conjugacy class has a fixed root and the same nontrivial quadratic character. Chebotarev gives a positive-density set of primes with a root of U_j and $p \equiv 2 \pmod 3$. Exclude the finitely many divisors of $\operatorname{disc}(U_j)$, $\operatorname{Res}(U_j, d)$, and $\operatorname{Res}(U_j, U_i)$ for $i \neq j$. All these integers are nonzero by irreducibility and unequal degrees. The asserted simple root and all other nonvanishing conditions follow. □

Theorem 6.4 (Prescribed exact multiplicities with periodic recurrence)**.** *Fix a finite window $I = P \sqcup C$, a positive integer M, and for each $j \in C$ a nonempty finite list of positive integers $e_{j,1}, \dots, e_{j,r_j}$. There are pairwise distinct primes $p_{j,k} \geq 5$ with $p_{j,k} \equiv 2 \pmod 3$, all coprime to M, and an arithmetic progression $\mathcal{A}$ of integer parameters with the following unconditional properties for every sufficiently large $a \in \mathcal{A}$:*

1. *$a \equiv -1 \pmod M$, and every odd-power trace of β_a is 1 modulo M;*
2. *for $i \in I$,*
$$v_{p_{j,k}}(U_i(a)) = \begin{cases} e_{j,k}, & i = j, \\ 0, & i \neq j; \end{cases}$$
3. *with*
$$T_{j,k} = \operatorname{ord}_{p_{j,k}^{e_{j,k}}(p_{j,k}^3 - 1)}(3),$$
one has
$$v_{p_{j,k}}(U_{j+hT_{j,k}}(a)) = e_{j,k} \qquad (h \geq 0).$$

Assuming hypothesis H, infinitely many $a \in \mathcal{A}$ additionally make every $U_i(a)$, $i \in P$, prime. Every prescribed composite position j then has at least r_j different prime divisors, with exactly the specified multiplicities for the chosen divisors. If P is empty, no conjecture is required.

Proof. For each pair (j, k) choose a fresh prime $p \geq 5$ and simple forcing root (p, b) from Lemma 6.3. Hensel's lemma lifts b uniquely to a root modulo p^{e+1}, where $e = e_{j,k}$. Among the p lifts of the root modulo p^e, choose any of the other $p - 1$ classes modulo p^{e+1}. Call it c. Taylor expansion and $U_j'(b) \neq 0$ give $v_p(U_j(c)) = e$; every other designated $U_i(c)$ is a unit modulo p. CRT now imposes $a \equiv -1 \pmod M$ and $a \equiv c_{j,k} \pmod{p_{j,k}^{e_{j,k}+1}}$ simultaneously. The resulting progression has modulus

$$D = M \prod_{j \in C} \prod_{k=1}^{r_j} p_{j,k}^{e_{j,k}+1}.$$

This proves the exact multiplicities within the finite window.

We verify recurrence at the same precision. Fix a chosen p, e and consider

$$\mathcal{B} = (\mathbb{Z}/p^{e+1}\mathbb{Z})[Z]/(f_a(Z)).$$

Because $p \nmid d(a)$, its reduction is etale. The square discriminant forces splitting type $1+1+1$ or 3, so every residue-field unit is killed by $p^3 - 1$. The kernel of reduction on $\mathcal{B}^\times$ is $1 + p\mathcal{B}$ and is killed by p^e, by induction with the binomial theorem. Hence every unit of $\mathcal{B}$ is killed by $E = p^e(p^3 - 1)$. The root transformation defining s has invertible denominator: a simultaneous zero of $1-QZ$ and $aZ+1$ would force $(a+1)^2 = 0 \bmod p$, excluded by $d(a) \neq 0$. Thus $\beta = -r^5 s$ is a unit in this algebra, and its traces agree with the integer polynomials. Since $p \equiv 2 \pmod 3$, $\gcd(E, 3) = 1$. For $T = \mathrm{ord}_E(3)$,

$$U_{n+T}(a) \equiv U_n(a) \pmod{p^{e+1}} \qquad (n \geq 0).$$

The nonzero residue of $U_j(a)/p^e$ modulo p is therefore retained at every $j + hT$, proving the exact recurring valuations.

Write the CRT progression as $a = A_0 + Dz$. For each $i \in P$, the polynomial $U_i(A_0 + Dz)$ is irreducible, primitive, and has positive leading coefficient. Their product has no fixed prime divisor: primes dividing M give values one; each forcing prime gives nonzero values at every prime-designated index; at a prime not dividing D choose z so that $A_0 + Dz \equiv -1$. Hypothesis H consequently supplies infinitely many simultaneous prime values. All other claims are unconditional congruences. Finally, every odd trace is an integer polynomial in a and has value one at $a = -1$, proving the modulo-M assertion. Taking a large ensures the Pisot bounds and makes each forced value larger than its chosen prime divisors. □

The theorem prescribes valuations of selected divisors, not a complete prime factorization: unselected prime factors may also occur. It strengthens the finite-pattern statement in the archive while retaining its essential finite-window quantifier and its explicit conjectural prime-value assumption.

6.4 Audit boundaries and remaining arithmetic question

The single-block norm-minus-one family of round 22 also has all-index local irreducibility, but its index-zero trace is a fourth power; its prime-prefix statement deliberately begins at the next index. The cyclic unit family above is stronger for conditional prime-prefix existence and starts at index zero. Round 23 is compatible with the present recurrence theorem: its isolated divisors are chosen at primes congruent to one modulo three, while the recurring divisors above lie at primes congruent to two modulo three. In round 23 the prime and unit may depend on the isolated index; the result does not say that every trace has such a divisor in one fixed tower.

Round 26 proves a separate polynomial-family obstruction: fixed nonzero norm and square discriminant force the polynomial coefficient functions to be algebraically dependent and, over $\mathbb{C}$, to factor through one common polynomial. Its logarithmic two-form argument and the normalization-to-an-affine-line argument are valid; they do not prevent the one-parameter construction used here. We do not use a conjectural cyclic lift of the round 22 family, the multivariate Hilbert construction of round 24, or an unverified primality certificate from round 25 as inputs to the proofs above.

Uniform irreducibility is an unconditional result already in the archive. Arbitrarily long prime prefixes remain conditional on H; exact prescribed local divisibility is unconditional. None implies one fixed algebraic number whose entire infinite trace tower is prime. Nor does any identify the least-prime path defining Mills' constant. The full cubic Pisot compositeness problem and unconditional Mills transcendence remain open within these arguments.

7 Finite local conditions and logarithmic unit lattices

The new non-Mills material does not supply an additional proved route to Mills transcendence. A useful transfer is nevertheless possible: the unit-lattice viewpoint gives a quantitative explanation of why finitely many local conditions and Saito's strict conjugate bounds can coexist. It identifies a genuine limitation of that combination of tests without assuming that any trace is prime.

Let K be a totally real cubic field, with real embeddings $\sigma_1, \sigma_2, \sigma_3$. Fix a unit $w \in \mathcal{O}_K^\times$ with

$$\sigma_1(w) > 0, \qquad \sigma_2(w) < 0, \qquad \sigma_3(w) > 0.$$

Its norm is -1. Let $\mathfrak{m}$ be a nonzero integral ideal, and set

$$E^+(\mathfrak{m}) = \{u \in \mathcal{O}_K^\times : \sigma_i(u) > 0 \ (1 \le i \le 3),\ u \equiv 1 \pmod{\mathfrak{m}}\}.$$

Its two-coordinate logarithmic lattice is

$$\Lambda_{\mathfrak{m}} = \{(\log \sigma_1(u), \log \sigma_2(u)) : u \in E^+(\mathfrak{m})\} \subset \mathbb{R}^2.$$

Write $\Delta_{\mathfrak{m}} > 0$ for its covolume in these coordinates. This convention avoids an extraneous factor of $\sqrt{3}$ from embedding the lattice into the plane $x + y + z = 0$ in $\mathbb{R}^3$.

Theorem 7.1 (Quantitative persistence in every admissible unit coset)**.** *Fix $0 < a < 1/2$. For $H > 1$, let $B_{w,\mathfrak{m},a}(H)$ count the units $\beta \in wE^+(\mathfrak{m})$ such that, on writing*

$$B = \sigma_1(\beta), \qquad r = \sigma_2(\beta), \qquad s = \sigma_3(\beta),$$

one has

$$1 < B \le H, \qquad 0 < s < -r < 1, \qquad -r < B^{-a}.$$

Then

$$B_{w,\mathfrak{m},a}(H) = \frac{1-2a}{4\Delta_{\mathfrak{m}}}(\log H)^2 + O_{K,w,\mathfrak{m},a}(\log H + 1). \tag{8}$$

Every counted B is a cubic Pisot number of norm -1. In addition,

$$-r < s^{a/(1-a)}, \qquad \beta \equiv w \pmod{\mathfrak{m}}.$$

In particular, for $a = 17/40$, the two strict bounds

$$-r < \min(B^{-17/40}, s^{17/23})$$

hold for quadratically many units in logarithmic height in every such coset.

Proof. The subgroup $E^+(\mathfrak{m})$ has finite index in the unit group: positivity imposes only a finite sign condition, and reduction modulo $\mathfrak{m}$ maps units to the finite group $(\mathcal{O}_K/\mathfrak{m})^\times$. Dirichlet's unit theorem therefore makes $\Lambda_{\mathfrak{m}}$ a full lattice of rank two. Positivity removes its possible torsion, so its logarithm map is injective.

The logarithms of the coset $wE^+(\mathfrak{m})$ form the translate

$$(\log \sigma_1(w), \log|\sigma_2(w)|) + \Lambda_{\mathfrak{m}}.$$

For a point of this translate put $x = \log B$ and $y = \log(-r)$. Because $|\operatorname{N}\beta| = 1$, one has $\log s = -x - y$. The asserted inequalities are exactly

$$0 < x \le \log H, \qquad -x/2 < y < -ax. \tag{9}$$

Indeed $y > -x/2$ is $-r > s$, and $y < -ax < 0$ gives $-r < 1$; then $-x - y < -x/2 < 0$, so $s < 1$ as well. The triangle in (9) has area

$$\int_0^{\log H} (1/2 - a)x\,dx = \frac{1-2a}{4}(\log H)^2$$

and boundary length $O_a(\log H)$. Counting translates of a fixed fundamental parallelogram gives its area divided by $\Delta_{\mathfrak{m}}$, with error $O(\log H + 1)$: only cells meeting a fixed-width neighborhood of the boundary can contribute to the error. Strict versus non-strict boundaries changes the count by at most the same order. This proves (8).

The dominant embedding exceeds one and the other two have modulus less than one, so B is Pisot. It cannot be rational, since a rational unit is ± 1; its degree is therefore three. Finally $B(-r)s = 1$, and the inequality $-r < B^{-a}$ is equivalent to $(-r)^{1-a} < s^a$, proving the additional bound. The congruence is the one defining the coset. □

Corollary 7.2 (Persistence of every fixed finite trace sieve)**.** *Take $\mathfrak{m} = M\mathcal{O}_K$ with integer $M \geq 2$. Every unit counted in Theorem 7.1 satisfies*

$$\mathrm{Tr}_{K/\mathbb{Q}}(\beta^j) \equiv \mathrm{Tr}_{K/\mathbb{Q}}(w^j) \pmod{M} \qquad (j \geq 1).$$

Thus, if a seed w has $\gcd(\mathrm{Tr}(w^{3^n}), M) = 1$ at every level $n \geq 0$, then all the counted balanced Pisot units have the same avoidance property at every level.

Proof. From $\beta - w \in M\mathcal{O}_K$ one obtains $\beta^j - w^j \in M\mathcal{O}_K$ for every positive integer j. The trace of an algebraic integer is an integer, so tracing gives the stated congruence. The last assertion follows separately at every level; no primality assertion is inferred from coprimality with M. □

The seed hypothesis in this corollary is essential. The theorem does not assert that an arbitrary finite sieve has an avoiding seed in a specified field. Instead it proves that, once a finite residue class is admissible, the archimedean restrictions alone cannot eliminate it: there are asymptotically $c(\log H)^2$ members satisfying the strict conjugate bounds. The regulator enters only through the positive covolume $\Delta_{\mathfrak{m}}$ and changes the leading constant, not the existence of these candidates. This is stronger than an unquantified ray construction but does not close the remaining norm-minus-one cyclic Pisot case.

Example 7.3 (An explicit nonempty congruence coset). Let r be the largest root of $X^3 - 25X^2 - 4X + 1$ and order its other roots as $s < 0 < t$. The rational root test excludes ± 1, the discriminant is 273^2, and root brackets give

$$25 < r < 26, \qquad -0.294 < s < -0.292, \qquad 0.135 < t < 0.137.$$

Thus $K = \mathbb{Q}(r)$ is cyclic cubic and r has signs $(+, -, +)$ and norm -1. Direct root enumeration of the polynomial over $\mathbb{F}_2, \mathbb{F}_5, \mathbb{F}_{17}$ shows it irreducible in all three cases; these primes are inert. The integral element $v = (r^2 + 2)/3$ satisfies

$$v^3 - 213v^2 + 290v - 99 = 0.$$

This polynomial reduces to $X(X-1)(X+1)$ modulo three, so 3 splits completely. The residues of r at its three primes are $(-1, 1, 1)$ in some order, as follows from its norm -1 and trace $25 \equiv 1 \pmod 3$.

Put $w = r^{66619}$, where $66619 = 7 \cdot 31 \cdot 307$. At 2, the order of r divides 7; at 5 and 17, the finite-field norm identities give $r^{31} = -1$ and $r^{307} = -1$, respectively. The relevant quotients of 66619 are odd. Consequently, for every odd integer $j \geq 1$,

$$\mathrm{Tr}(w^j) \equiv \begin{cases} 1 & \pmod 2, \\ 1 & \pmod 3, \\ 2 & \pmod 5, \\ 14 & \pmod{17}. \end{cases}$$

Applying Theorem 7.1 to $wE^+(510\mathcal{O}_K)$ gives asymptotically $3(\log H)^2/(80\Delta_{510\mathcal{O}_K})$ strictly Saito-balanced Pisot units whose entire cubic-power trace towers avoid $2, 3, 5, 17$. This asserts finite-prime avoidance, not primality.

8 A complete cyclic cubic norm class

Submitted round S14 already proves that an eventually prime, growing trace tower in a cyclic cubic field must have norm congruent to zero or minus one modulo three [2, Theorem 1]. Thus the norm class excluded below is not newly discovered here. Combining that restriction with the trace-limit classification and cyclic Frobenius-return argument of [15], we record a positive-density formulation for the actual Pisot floors, including both possible real floor branches. This is a synthesis of existing ingredients, not a new return mechanism or a claim of historical priority.

Theorem 8.1 (Cyclic cubic norms congruent to one)**.** *Let $K/\mathbb{Q}$ be a cyclic cubic extension, and let $\beta \in K$ be a cubic Pisot number. If*

$$N = \mathrm{N}_{K/\mathbb{Q}}(\beta) \equiv 1 \pmod 3,$$

then the set

$$\mathcal{C}_\beta = \{n \geq 0 : \lfloor \beta^{3^n} \rfloor \text{ is composite}\}$$

has positive lower density. In particular, this conclusion holds for every cyclic cubic Pisot unit of norm $+1$.

Proof. A cyclic cubic field is totally real, since a nontrivial complex conjugation would have order two in a group of order three. Write the conjugates of β as $\beta, \gamma_2, \gamma_3$, and put

$$U_n = \mathrm{Tr}_{K/\mathbb{Q}}(\beta^{3^n}), \qquad f(X) = X^3 - uX^2 + vX - N.$$

The two small conjugates have different absolute values: equality would give $\gamma_2 = -\gamma_3$ and hence $\beta = u \in \mathbb{Z}$, contrary to degree three. Thus their odd-power sum has a fixed nonzero sign. For all sufficiently large n,

$$\lfloor \beta^{3^n} \rfloor = U_n - s, \qquad s \in \{0, 1\},$$

where $s = 1$ exactly when the small conjugate of larger absolute value is positive. Also $U_n \to +\infty$. Let $\lambda = \lim_{n\to\infty} U_n$ in $\mathbb{Z}_3$.

If $s = 1$, Proposition 2.4 and its explicit table show that either equality $\lambda = 0$ or $\lambda = 2$ would force $3 \mid N$. Since $N \equiv 1 \pmod 3$, neither equality is possible. Theorem 2.2, with the single shift $h = -1$, supplies an eventual arithmetic progression of composite floors.

Suppose now that $s = 0$. If $\lambda \notin \{1, -1\}$, Theorem 2.2 with $h = 0$ gives the same conclusion. We handle the two remaining limits separately.

If $\lambda = 1$, the residue classification gives

$$f(X) \equiv (X - 1)(X^2 + 1) \pmod 3,$$

because $N \equiv 1 \pmod 3$. Its root 1 is simple, so Hensel's lemma gives an embedding $K = \mathbb{Q}(\beta) \hookrightarrow \mathbb{Q}_3$. In a Galois extension all primes over three have the same local degree; the degree-one completion therefore implies that three splits completely in K. The three conjugates of β now lie in $\mathbb{Q}_3$ and are units, because $3 \nmid N$. Each has Teichmüller part 1 or -1. Their sum is $\lambda = 1$, so these three parts must be $1, 1, -1$ in some order. Their product gives $N \equiv -1 \pmod 3$, a contradiction. Equivalently, the displayed separable reduction has factorization type $1 + 2$, which cannot occur at an unramified prime in a cyclic cubic field.

It remains to treat $\lambda = -1$. Lemma 2.1 gives $U_n \equiv -1 \pmod 3$ for every n. If only finitely many U_n are prime, then every sufficiently late floor is composite, and the density is one. Otherwise choose a sufficiently late prime $p = U_r$ with

$$p \nmid 3N \operatorname{disc}(f).$$

This is possible since $U_n \to +\infty$. Frobenius in the cyclic cubic field has cycle type $1+1+1$ or 3. Consequently all roots of f modulo p lie in $\mathbb{F}_{p^3}$ and are nonzero. Since $p \equiv -1 \pmod 3$,

$$\gcd(3, p^3 - 1) = 1.$$

Choose $D \geq 1$ with $3^D \equiv 1 \pmod{p^3-1}$. For every residual root a and every $j \geq 0$ this gives $a^{3^{r+jD}} = a^{3^r}$. Summing over the three roots yields

$$U_{r+jD} \equiv U_r = 0 \pmod p \qquad (j \geq 0).$$

For sufficiently large j the floor equals $U_{r+jD} > p$, so this is an arithmetic progression of composite floors. Every progression obtained above has positive lower density, completing the proof. □

Remark 8.2 (What remains in the cyclic Mills branch). For a hypothetical cyclic cubic shadow of the least Mills constant, the dominant small conjugate is negative. An eventually prime trace tail therefore forces $\lambda = 1$: other limits are excluded by Theorem 2.2 or by the last part of the proof above. If its norm is prime to three, complete splitting at three then forces $N \equiv -1 \pmod 3$. In particular, a unit shadow must have norm -1. The norm--1 cyclic units constructed elsewhere in this paper show why Theorem 8.1 does not settle the entire cyclic case or unconditional Mills transcendence.

9 Excluding rational powers of Mills' constant from number fields

Restricted cubic results already give complete exclusion statements for other, substantially larger classes of algebraic expressions. Throughout this section A is the least Mills constant defined in the introduction. The cubic three-adic embedding obstruction below was already proved in the supplied manuscript [16]. We include a short proof from Theorem 2.2 and Proposition 2.4 to make the subsequent field deductions self-contained. The ambient-field and cyclotomic consequences are deductions from this obstruction; no historical priority is asserted for them.

Lemma 9.1 (Prime traces force a three-adic embedding). *Let β be a cubic Pisot number, with minimal polynomial $f(X) = X^3 - uX^2 + vX - N$, and put $U_n = \operatorname{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n})$. If U_n is prime for every sufficiently large n, then its three-adic limit satisfies $\lambda \in \{1, -1\}$, and $\mathbb{Q}(\beta)$ admits an embedding into $\mathbb{Q}_3$. No restriction on the norm N is required.*

Proof. The Pisot property gives $U_n = \beta^{3^n} + o(1)$, so $U_n \to \infty$. If $\lambda \notin \{1, -1\}$, apply Theorem 2.2 with the single shift 0. It gives infinitely many composite U_n, contrary to the hypothesis. Write $\lambda = \varepsilon$, where $\varepsilon \in \{1, -1\}$. The six corresponding coefficient classes in Proposition 2.4 satisfy

$$u \equiv \varepsilon, \qquad N \equiv \varepsilon v \pmod 3, \qquad f(X) \equiv (X - \varepsilon)(X^2 + v) \pmod 3.$$

If $v \not\equiv 2 \pmod 3$, the residual root ε is simple. If $v \equiv 2 \pmod 3$, the residual polynomial is $(X - \varepsilon)^2(X + \varepsilon)$, whose root $-\varepsilon$ is simple. Hensel's lemma therefore gives a root of f in $\mathbb{Z}_3$. Since f is irreducible over $\mathbb{Q}$, evaluation at this root embeds $\mathbb{Q}[X]/(f) = \mathbb{Q}(\beta)$ into $\mathbb{Q}_3$. □

Lemma 9.2 (Pisot powers retain their field). *If β is a Pisot number and r is a positive integer, then $\mathbb{Q}(\beta^r) = \mathbb{Q}(\beta)$.*

Proof. An embedding of $\mathbb{Q}(\beta)$ into $\mathbb{C}$ that fixes $\mathbb{Q}(\beta^r)$ sends β to a conjugate γ with $\gamma^r = \beta^r$. Hence $|\gamma| = \beta > 1$. The defining Pisot condition leaves only $\gamma = \beta$. Separability in characteristic zero says that the number of such embeddings is $[\mathbb{Q}(\beta) : \mathbb{Q}(\beta^r)]$, which is therefore one. □

Theorem 9.3 (An obstruction in every ambient number field)**.** *Let $K \subseteq \mathbb{C}$ be a number field. If $A^q \in K$ for some $q \in \mathbb{Q} \setminus \{0\}$, then K contains a totally real cubic subfield F admitting an embedding into $\mathbb{Q}_3$. Consequently, if K has no such subfield, then*

$$\boxed{A^q \notin K \qquad (q \in \mathbb{Q} \setminus \{0\}).}$$

Here rational powers of $A > 1$ denote their positive real values.

Proof. Replacing q by $-q$ if necessary, write $q = r/s$ with positive integers r, s. The inclusion $A^{r/s} \in K$ makes A algebraic. Saito's reduction [5, Theorem 1.7] supplies $\beta = A^{3^m}$, a totally real cubic Pisot number satisfying (1). Its two small conjugates have negative sum at every sufficiently large odd power: the negative one has strictly larger absolute value. Thus

$$\lfloor \beta^{3^n} \rfloor = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n})$$

for every sufficiently large n. These integers are prime by the definition of A. Lemma 9.1 therefore gives a $\mathbb{Q}_3$-embedding of $F = \mathbb{Q}(\beta)$. Finally,

$$\beta^r = (A^{r/s})^{s3^m} \in K, \qquad \mathbb{Q}(\beta^r) = F$$

by Lemma 9.2, proving $F \subseteq K$. □

Corollary 9.4 (Cubic fields and radical expressions)**.** *If a cubic number field K has no embedding into $\mathbb{Q}_3$, then no nonzero rational power of A belongs to K. In particular this holds when 3 is inert or totally ramified in K. Equivalently, no positive element of such a field, raised to a nonzero rational power, can equal A.*

Proof. Any cubic subfield of the cubic field K is K itself, so Theorem 9.3 applies. Inertness and total ramification both exclude a prime of K over 3 with ramification index and residue degree equal to one, which an embedding into $\mathbb{Q}_3$ would require. □

Theorem 9.5 (A Galois prime-count obstruction)**.** *Let $K/\mathbb{Q}$ be a finite Galois extension and let $g_3(K)$ be the number of prime ideals of K above 3. If $3 \nmid g_3(K)$, then $A^q \notin K$ for every $q \in \mathbb{Q} \setminus \{0\}$.*

Proof. Suppose otherwise and let $F \subseteq K$ be the cubic subfield from Theorem 9.3. Put $G = \mathrm{Gal}(K/\mathbb{Q})$ and $H = \mathrm{Gal}(K/F)$, so $[G : H] = 3$. The $\mathbb{Q}_3$-embedding supplies a prime $\mathfrak{p}$ of F with completion $F_\mathfrak{p} = \mathbb{Q}_3$. Choose a prime $\mathfrak{P}$ of K above $\mathfrak{p}$, and let $D \subseteq G$ be its decomposition group. The image of F in $K_\mathfrak{P}$ lies in $\mathbb{Q}_3$, so every member of D fixes F and $D \subseteq H$. The usual transitive action on the primes above 3 gives

$$g_3(K) = [G : D] = [G : H][H : D] = 3[H : D],$$

contrary to the hypothesis. □

Remark 9.6 (The exact abelian scope of the local obstruction). If $K/\mathbb{Q}$ is finite abelian, it contains a cyclic cubic subfield in which 3 splits completely if and only if $3 \mid g_3(K)$. Indeed, its decomposition group D is normal. An index-three subgroup H containing D exists precisely when the finite abelian group G/D has order divisible by three. Its fixed field is cyclic cubic, is totally real because its degree is odd, and has trivial decomposition group at 3. This equivalence concerns the existence of a subfield with the required local property; it does not assert that a Mills candidate exists in that subfield.

Corollary 9.7 (Cyclotomic exclusions)**.** *Write $N = 3^b m$, where $b \geq 0$ and $(m, 3) = 1$. If*

$$3 \nmid \frac{\varphi(m)}{\operatorname{ord}_m(3)},$$

then $A^q \notin \mathbb{Q}(\zeta_N)$ for every nonzero rational q. Here ζ_N is a primitive Nth root of unity, and $\varphi(1) = \operatorname{ord}_1(3) = 1$ by convention. In particular, for every nonzero rational q,

$$A^q \notin \bigcup_{a,b\geq 0} \mathbb{Q}(\zeta_{7^a 3^b}), \qquad A^q \notin \bigcup_{a,b\geq 0} \mathbb{Q}(\zeta_{13^a 3^b}).$$

Proof. In $\mathbb{Q}(\zeta_m)$ the prime 3 is unramified and its Frobenius has order $\operatorname{ord}_m(3)$, so it has $\varphi(m)/\operatorname{ord}_m(3)$ primes above it. The 3-power cyclotomic factor is totally ramified at 3. Locally, adjoining this totally ramified factor to the unramified completion multiplies the local degree by $\varphi(3^b)$, just as the global degree is multiplied by $\varphi(3^b)$. Thus it does not change the number of primes:

$$g_3(\mathbb{Q}(\zeta_{3^b m})) = \frac{\varphi(m)}{\operatorname{ord}_m(3)}.$$

Theorem 9.5 proves the first assertion.

For $a \geq 1$, elementary lifting of multiplicative orders gives

$$\operatorname{ord}_{7^a}(3) = 6 \cdot 7^{a-1}, \qquad \operatorname{ord}_{13^a}(3) = 3 \cdot 13^{a-1}.$$

Indeed, the orders modulo 7 and 13 are 6 and 3, respectively, and $v_7(3^6 - 1) = v_{13}(3^3 - 1) = 1$. The corresponding values of g_3 are 1 and 4, neither divisible by three. The case $a = 0$ has $g_3 = 1$. Every element of either displayed union lies in one finite stage, proving the two exclusions. □

Corollary 9.8 (A quotient obstruction)**.** *Let $K/\mathbb{Q}$ be finite Galois. If $\operatorname{Gal}(K/\mathbb{Q})$ has no quotient isomorphic to C_3 or S_3, then $A^q \notin K$ for every nonzero rational q. This applies, in particular, when the Galois group is perfect, or is S_d with $d \geq 5$.*

Proof. If a cubic subfield F existed, its normal closure would lie in K and have a transitive cubic Galois group, necessarily C_3 or S_3. Restriction would give the prohibited quotient. Apply Theorem 9.3. Quotients of a perfect group are perfect, whereas C_3 and S_3 are not. For $d \geq 5$, the only proper nontrivial quotient of S_d is C_2. □

Remark 9.9 (Composita and the remaining cubic problem)*.* These exclusions cannot be combined by simply taking composita. For example, $\mathbb{Q}(\zeta_7)$ and $\mathbb{Q}(\zeta_{13})$ separately satisfy Theorem 9.5, but their compositum has

$$g_3(\mathbb{Q}(\zeta_{91})) = \frac{72}{\operatorname{lcm}(6, 3)} = 12.$$

By Remark 9.6, that compositum contains a cyclic cubic subfield in which 3 splits completely.

The section excludes entire fields and all rational powers at once, without a unit or norm hypothesis. Nevertheless, it does not eliminate the cubic shadows already satisfying $\lambda = \pm 1$, a $\mathbb{Q}_3$-embedding, and Saito's real-conjugate bounds. In the cyclic case such a shadow must have 3 completely split. Excluding every remaining Mills shadow would suffice for unconditional transcendence of A, even without solving the stronger question for every cubic Pisot number. No such final exclusion is obtained here.

10 The exact field boundary for a fixed finite set of prime divisors

The unit-coset result above assumes an admissible seed. Allowing a fixed nonzero integral norm, rather than requiring norm minus one, removes that assumption in every remaining totally real cubic field. The resulting statement concerns divisibility by a fixed finite set of primes; it makes no assertion that the surviving integers are prime.

For a cubic algebraic integer β, write

$$\lambda(\beta) = \lim_{n\to\infty} \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3^n}) \quad \text{in } \mathbb{Z}_3.$$

Its existence and its dependence on the residual polynomial are the three-adic convergence and classification in Proposition 2.4 and its preceding construction.

Theorem 10.1 (Every admissible field survives every fixed finite prime sieve)**.** *Let K be a totally real cubic field, with real embeddings $\sigma_1, \sigma_2, \sigma_3$. The following are equivalent.*

1. *There is a field embedding $K \hookrightarrow \mathbb{Q}_3$.*
2. *For every finite set P of rational primes, there are infinitely many $\beta \in \mathcal{O}_K$ such that, on writing*

$$B = \sigma_1(\beta), \qquad \sigma_2(\beta) = -\rho, \qquad \sigma_3(\beta) = s,$$

one has

$$B > 1, \qquad 0 < s < \rho < 1, \qquad \rho < \min\{B^{-17/40}, s^{17/23}\}, \tag{10}$$

$$\lambda(\beta) = 1, \qquad \gcd\left(\lfloor B^m \rfloor, \prod_{p\in P} p\right) = 1 \quad \textit{for every positive odd integer } m. \tag{11}$$

Here every B in (2) is a cubic Pisot number generating K. For fixed K, P, the elements in (2) may be chosen in one coset $\alpha E^+(M\mathcal{O}_K)$, where

$$M = \prod_{p\in P\cup\{3\}} p, \qquad \mathrm{N}_{K/\mathbb{Q}}(\alpha) = -N, \qquad N \geq 1, \qquad \gcd(N, M) = 1.$$

Thus their norm is the same integer $-N$. With Δ_M denoting the covolume of the logarithmic lattice of $E^+(M\mathcal{O}_K)$ in its first two embedding coordinates, the number of these elements satisfying (10) and $B \leq H$ is

$$\frac{3}{80\Delta_M}(\log H)^2 + O_{K,\alpha,M}(\log H + 1). \tag{12}$$

The entire statement remains true with $\lambda(\beta) = -1$.

Proof. Assume first that $K \hookrightarrow \mathbb{Q}_3$. Then the degree-three étale algebra $K \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is either $\mathbb{Q}_3^3$ or $\mathbb{Q}_3 \times E$, with $E/\mathbb{Q}_3$ a quadratic field. The corresponding decomposition of $\mathcal{O}_K \otimes \mathbb{Z}_3$ allows the following prescriptions modulo three.

Local algebra	Prescribed residues	Residual polynomial
$\mathbb{Q}_3^3$	$(1, 1, -1)$	$(X-1)^2(X+1)$
$\mathbb{Q}_3 \times E$, E ramified	$(-1, 1)$	$(X+1)(X-1)^2$
$\mathbb{Q}_3 \times E$, E unramified	$(1, i)$, $i^2 = -1$ in $\mathbb{F}_9$	$(X-1)(X^2+1)$

In the ramified case the residue 1 is the scalar element of $\mathcal{O}_E/3\mathcal{O}_E$, so its multiplication operator has characteristic polynomial $(X-1)^2$, including multiplicity. In the unramified case $i \notin \mathbb{F}_3$, and its two residual conjugates are i and $-i$. Every listed element is a unit modulo three.

For each $p \in P \setminus \{3\}$ prescribe the scalar residue $1 \in \mathcal{O}_K/p\mathcal{O}_K$. The Chinese remainder theorem gives a nonempty coset $\alpha_0 + M\mathcal{O}_K$ satisfying all these prescriptions. Under the three real embeddings, $M\mathcal{O}_K$ is a full lattice in $\mathbb{R}^3$. Every translate of such a lattice meets the open orthant $(+,-,+)$: a sufficiently large multiple of a vector in that orthant has a lattice-coset point within a fixed distance. Choose α in this intersection. Its norm is $-N$ for a positive integer N, and all prescribed residues are units, giving $\gcd(N, M) = 1$.

Every element $\beta \in \alpha E^+(M\mathcal{O}_K)$ has the same signs, norm, and residues as α. Consequently, for every positive integer m and every $p \in P \setminus \{3\}$,

$$\operatorname{Tr}(\beta^m) \equiv 3 \pmod{p}.$$

For every positive odd m, the three-adic prescriptions instead give $\operatorname{Tr}(\beta^m) \equiv 1 \pmod 3$: the residual roots are either $1, 1, -1$ or $1, i, -i$, with multiplicity. Their Teichmüller lifts have sum one, so the classification in Proposition 2.4 gives $\lambda(\beta) = 1$ whenever β generates K.

It remains to count the elements with the required real conjugates. The subgroup $E^+(M\mathcal{O}_K)$ has finite index in the unit group. By Dirichlet's unit theorem [1, Chapter 5], its first two logarithmic coordinates therefore form a full rank-two lattice of covolume Δ_M. Positivity removes the torsion kernel. The corresponding coordinates of $\alpha E^+(M\mathcal{O}_K)$ form a translate of this lattice. Put $x = \log B$, $y = \log \rho$, and $L = \log N \geq 0$. The norm identity gives

$$\log s = L - x - y.$$

For $a = 17/40$, the inequalities $1 < B \leq H$, $0 < s < \rho < 1$, and $\rho < B^{-a}$ are exactly

$$0 < x \leq \log H, \qquad \frac{L-x}{2} < y < -ax. \tag{13}$$

This region has area

$$\frac{1-2a}{4}(\log H)^2 + O_N(\log H + 1)$$

and boundary length $O_N(\log H + 1)$. Counting fundamental parallelograms, as in Theorem 7.1, gives (12). The leading constant is positive.

For every counted point, the dominant embedding exceeds one and the other two have modulus less than one. Thus B is Pisot and cannot be rational; because $[K : \mathbb{Q}] = 3$, it generates K. The additional bound in (10) follows from the integer norm:

$$\rho^{40} < B^{-17} \leq N^{17}B^{-17} = (\rho s)^{17}, \qquad \text{so} \qquad \rho < s^{17/23}.$$

Finally, if m is positive and odd, then $-\rho^m + s^m \in (-1, 0)$. Therefore

$$\lfloor B^m \rfloor = \operatorname{Tr}_{K/\mathbb{Q}}(\beta^m),$$

and the preceding residual calculations prove (11).

Conversely, a single element from (2) generates K and has $\lambda = 1$. Proposition 2.4 shows that its minimal polynomial modulo three is one of

$$X^2(X-1), \qquad (X-1)(X^2+1), \qquad (X-1)^2(X+1).$$

Each has a simple linear root over $\mathbb{F}_3$. Hensel's lemma lifts that root to $\mathbb{Q}_3$, and the resulting root of the irreducible minimal polynomial defines an embedding $K \hookrightarrow \mathbb{Q}_3$.

For the variant with $\lambda = -1$, negate only the prescribed residues at three. The real sign choice is still made independently in the same open orthant. The odd traces are now -1 modulo three, and the Teichmüller sum is -1; all other arguments are unchanged. The reverse implication follows by negating the three residual polynomials in the preceding paragraph. □

Corollary 10.2 (Normalized survivors in every real interval)**.** *Fix an admissible field K, a finite prime set P, and a nonempty open interval $I \subset (1,\infty)$. For every sufficiently large integer k, there are elements β from Theorem 10.1 such that*

$$B^{1/3^k} \in I, \qquad \gcd\left(\left\lfloor (B^{1/3^k})^{3^{k+n}} \right\rfloor, \prod_{p\in P} p\right) = 1 \quad (n \geq 0).$$

If I is contained in an open interval on which a prescribed finite list of prime floors $\lfloor t^{3^j} \rfloor$ is constant, these normalized survivors have that same finite prime prefix.

Proof. Choose $1 < a < b$ with $(a,b) \subset I$. Subtracting (12) at $H = b^{3^k}$ and $H = a^{3^k}$ gives

$$\frac{3}{80\Delta_M} 3^{2k}((\log b)^2 - (\log a)^2) + O_{K,\alpha,M,a,b}(3^k),$$

which is positive for all sufficiently large k. Any resulting $a^{3^k} < B \leq b^{3^k}$ can be obtained with strict upper endpoint as well, by choosing a smaller inner interval. The displayed tail identity is simply $(B^{1/3^k})^{3^{k+n}} = B^{3^n}$. The assertion about the finite prefix follows from membership in I. □

Remark 10.3 (Scope of the obstruction)*.* The quantifiers in Theorem 10.1 are

$$\text{for every finite } P, \text{ there exists a fixed } N \text{ and infinitely many } \beta.$$

The integer N and the coset may depend on P; no one element is asserted to avoid every prime. The normalized construction likewise asserts neither an all-prime tail nor a prime condition at the intervening indices between a fixed prefix and its prescribed tail. What is ruled out is a field-uniform proof using only Saito's strict conjugate bounds, the limiting trace condition, and divisibility by one fixed finite set of primes. The theorem does not rule out adaptive choices of divisors depending on β, other arithmetic information, or an argument using the least-prime property specific to the minimum defining Mills' constant.

11 The remaining Mills alternative

The results yield several rigorous forms of progress, but each has a precise quantifier. The simultaneous-window theorem closes a residue-defined subclass. The coefficient sieve controls the size of a possible exceptional set in boxes. The Pell certificates exclude entire arithmetic classes of parameters in one family. The conditional construction realizes every prescribed *finite* pattern, and the unit-lattice theorem preserves *fixed finite* local data. The new residual-field theorem supplies its own seed in every three-adically admissible cubic field; this still preserves a finite-divisor quantifier, not primality. None of these statements exchanges a finite quantifier for an infinite one.

Three specific obstructions remain.

1. At a nonintegral three-adic limit, the prime-factor order budget works for every fixed shift. At $\lambda = \pm 1$, its key distance from ± 1 is zero for the unshifted trace. These are exactly the residual classes in which a new argument is required.

2. A coefficient set of stretched-exponentially small proportion can still contain a particular algebraic candidate. An ambient box estimate gives no automatic exclusion on a fixed field or on the exponentially sparse Pell orbit. The separate Pell calculation is needed precisely for this reason.

3. An infinite collection of algebraic numbers, each with a long finite prime prefix, need not contain one number with an infinite prime tower. Nor does it determine the minimum defining Mills' constant. The minimum requires a global prime-chain argument in addition to local polynomial information.

Consequently the remaining full targets are unchanged: Saito's cubic Pisot Problem 1.1, the exclusion of the Mills cubic alternative, and irrationality of the least square-clock constant. What is established here is a stronger collection of restricted theorems, quantitative bounds, and exact exclusions, with reusable proofs.

A concrete remaining target using minimality. For a surviving cubic candidate let $T_n = \mathrm{Tr}(\beta^{3^n})$. A hypothetical algebraic least Mills constant requires both eventual primality of T_n and, for every sufficiently large n,

$$T_{n+1} = \mathrm{nextprime}(T_n^3).$$

The latter property is an inherited observation from the supplied manuscript [010]. To see why it holds, the prime-interval bound used in [5] extends every sufficiently large prime q to a prime child strictly between q^3 and $(q+1)^3 - 1$. An earlier prime child of T_n could therefore be continued to an infinite nested prime chain below the defining minimum, a contradiction. Consequently it would suffice to prove the following restricted, still unproved assertion: every surviving candidate either has infinitely many composite T_n, or admits arbitrarily large n with a prime strictly between T_n^3 and T_{n+1}. This target uses the least-prime selection rule that the fixed-prime survival theorem does not encode. It would settle Mills transcendence without requiring composite floors for all cubic Pisot numbers.

Verification boundary. The central symbolic arguments were independently checked in parallel. The residual-field proofs were checked independently, including the ramified local algebra at three, all-rational-power quantifiers, and the distinction between a finite-prime sieve and an all-prime tower. Exact arithmetic replayed the two finite-field factorizations of the balanced seed, the old 26 second-orbit Pell witnesses and five first-orbit witnesses, the three new witnesses, and the completeness of the new residue cover. Additional exhaustive small-prime checks test the shifted local counts; they are consistency checks, not substitutes for the all-degree proof. No large numerical prime certificate from an earlier submission is needed for the new theorems. A compact standard-library certificate is included below.

A The inherited Pell reduction and the exact finite cover

For self-contained use of the new certificate we record the earlier reduction and cover. The substitutions $x = 8u + 5$, $y = 4w$ transform $4u^2 + 5u + 23 = 2w^2$ into

$$x^2 - 2y^2 = -343, \qquad x \equiv 5 \pmod 8, \quad y \equiv 0 \pmod 4.$$

Multiplying $x + y\sqrt{2}$ by powers of $3 - 2\sqrt{2}$ reduces it uniquely to $[\sqrt{343}, (3 + 2\sqrt{2})\sqrt{343})$. This bounds $0 < x < 53$, $0 < y < 40$. Enumeration gives exactly $(7, 14), (13, 16), (25, 22), (35, 28)$. Their four-term residue cycles modulo eight retain only the first seed at odd powers of $3 + 2\sqrt{2}$, and the second at even powers. These become the two orbits from $(9, 14)$ and $(1, 4)$ under T. The latter initial point is omitted because $u = 1$ is a reducible degeneration.

For the first orbit a_j, the residues have periods two, ten, and five modulo $3, 19, 29$ respectively:

3	$0, 1$
19	$9, 8, 17, 1, 18, 4, 5, 15, 12, 14$
29	$9, 12, 13, 15, 24.$

Even indices give trace zero modulo three. The five odd classes $j = 1, 3, 5, 7, 9 \pmod{10}$ are covered by $(p, c, n) = (29, 12, 41), (29, 15, 5), (19, 4, 17), (29, 13, 1), (19, 14, 7)$. Every listed zero is beyond its root-order return budget. For the second orbit odd j give trace zero modulo three. Writing $j = 2k$, the following old rows cover all k except $0, 2187 \pmod{6561}$; a pair (a, n) means a recurring zero at trace index n whenever $k \equiv a \pmod{m}$.

p	m	(a, n)
5	3	$(1, 3)$
7	3	$(2, 8)$
179	9	$(3, 52)$
199	9	$(6, 126)$
107	27	$(9, 4), (18, 25)$
487	81	$(27, 366), (54, 1333)$
4373	729	$(162, 2379), (486, 7323), (567, 3228)$
20411	729	$(81, 7662), (405, 34139)$
8747	2187	$(243, 7161), (324, 2814), (648, 105), (729, 830)$
		$(1053, 2330), (1701, 3456), (1782, 4165), (2106, 12465)$
52489	2187	$(1377, 54889)$
39367	6561	$(3645, 45679), (4374, 15588)$
629857	2187	$(972, 647491), (1458, 678551)$

The three new rows of Theorem 5.1 reduce the survivor set to $k \equiv 0 \pmod{19683}$.

To verify the valuation formula used there, set $(17 + 12\sqrt{2})^{2k} = C_k + D_k\sqrt{2}$. Then

$$b_{2k} - 1 = \frac{13(C_k - 1) + 32D_k}{8}, \quad D_{3k} = D_k(3 + 8D_k^2).$$

For $3 \nmid k$, reduction modulo nine gives $v_3(D_k) = 1$. Repeated cubing gives $v_3(D_k) = 1 + v_3(k)$ for every $k > 0$. Since $C_k^2 - 2D_k^2 = 1$ and $C_k \equiv 1 \pmod{3}$, $v_3(C_k - 1) = 2v_3(D_k)$. The term involving D_k therefore determines the numerator's valuation, proving $v_3(b_j - 1) = 1 + v_3(j)$ for even $j > 0$.

B A reproducible exact certificate

The following Python 3 code uses only the standard library. It checks primality by trial division, modular periods in the quadratic ring, cubic trace zeros by binary polynomial powering, and the full residue cover. It never constructs a trace of millions-digit exponent. The algebraic reasons why these finite checks imply infinite progressions are given in the proofs above.

```
from math import isqrt

rows=[(5,3,[(1,3)]),(7,3,[(2,8)]),(179,9,[(3,52)]),(199,9,[(6,126)]),(107,27,[(9,4),(18,25)])
     ,(487,81,[(27,366),(54,1333)]),(4373,729,[(162,2379),(486,7323),(567,3228)]),(20411,729,[(81,7662)
     ,(405,34139)]),(8747,2187,[(243,7161),(324,2814),(648,105),(729,830),(1053,2330),(1701,3456)
     ,(1782,4165),(2106,12465)]),(52489,2187,[(1377,54889)]),(39367,6561,[(3645,45679),(4374,15588)])
     ,(629857,2187,[(972,647491),(1458,678551)])]
def v3(n):
 a=0
 while n%3==0 and n: a+=1;n//=3
 return a

def qm(x,y,p):return ((x[0]*y[0]+2*x[1]*y[1])%p,(x[0]*y[1]+x[1]*y[0])%p)
def qp(x,n,p):
 r=(1,0)
 while n:
  if n&1:r=qm(r,x,p)
  x=qm(x,x,p);n//=2
 return r

def b(j,p):
```

```
 x,y=qm((13,16),qp((17,12),j,p),p)
 return ((x-5)*pow(8,-1,p))%p

def pm(a,b,u,p):
 c=[0]*5
 for i,x in enumerate(a):
  for j,y in enumerate(b):c[i+j]=(c[i+j]+x*y)%p
 for k in (4,3):
  v=c[k];c[k]=0;c[k-1]=(c[k-1]+u*v)%p;c[k-2]=(c[k-2]+v)%p;c[k-3]=(c[k-3]-v)%p
 return c[:3]
def pp(a,n,u,p):
 r=[1,0,0]
 while n:
  if n&1:r=pm(r,a,u,p)
  a=pm(a,a,u,p);n//=2
 return r

def trace(n,u,p):
 M=(p-1)*(p*p-1)*(p**3-1); E=pow(3,n,M); A,B,C=pp([0,1,0],E,u,p)
 return (3*A+u*B+(u*u+2)*C)%p
for p,m,items in rows:
 assert p>=2 and all(p%d for d in range(2,isqrt(p)+1)),p
 assert qp((17,12),2*m,p)==(1,0),(p,m)
 for a,n in items:
  assert n>=v3((p-1)*(p*p-1)*(p**3-1)),(p,n)
  assert trace(n,b(2*a,p),p)==0,(p,m,a,n,b(2*a,p))
surv=[k for k in range(6561) if all(all(k%m!=a for a,n in items) for p,m,items in rows)]
assert surv==[0,2187],surv
for p,c,n in [(29,12,41),(29,15,5),(19,4,17),(29,13,1),(19,14,7)]:
 assert trace(n,c,p)==0
 assert n>=v3((p-1)*(p*p-1)*(p**3-1))
print('All 26 second-orbit witnesses and 5 first-orbit witnesses verified by exact arithmetic.')
print('Surviving k residues modulo6561:',surv)

# Three additional witnesses: (prime, parameter period, class, trace index).
new=[(8279983,6561,2187,4863973),
     (27398737,19683,6561,4929815),
     (15037811,19683,13122,549606)]
for p,m,a,n in new:
 assert all(p%d for d in range(2,isqrt(p)+1))
 assert qp((17,12),2*m,p)==(1,0)
 assert n>=v3((p-1)*(p*p-1)*(p**3-1))
 assert trace(n,b(2*a,p),p)==0
remaining=[k for k in range(19683)
 if all(all(k%m!=a for a,n in items) for p,m,items in rows)
 and all(k%m!=a for p,m,a,n in new)]
assert remaining==[0]
# Reduced Pell seeds and first-orbit parameter periods.
seeds=[(x,y) for x in range(1,53) for y in range(1,40)
       if x*x-2*y*y==-343]
assert seeds==[(7,14),(13,16),(25,22),(35,28)]
def orbit(seed,p,length):
 u,w=seed;out=[]
 for j in range(length):
  out.append(u%p);u,w=(17*u+12*w+10)%p,(24*u+17*w+15)%p
 assert (u,w)==(seed[0]%p,seed[1]%p)
 return out
assert orbit((9,14),3,2)==[0,1]
assert orbit((9,14),19,10)==[9,8,17,1,18,4,5,15,12,14]
assert orbit((9,14),29,5)==[9,12,13,15,24]
assert orbit((1,4),3,2)==[1,0]
print('All certificates verified; surviving k modulo19683:',remaining)
```

C Source map and attribution

The supplied archive is cited as research material, not as a collection of independently published papers. Round identifiers below refer to `Race244_Topic02(1).zip`; S means Showa. The earlier numbered manuscripts refer to the previously supplied 36-paper collection.

Source	Use and status
S12, S27	Local factor blocks and uniform irreducibility, reproduced and audited.
S28	Finite prime/composite patterns under H; exact multiplicities are added here.
S35, S36	Complete Pell reduction and inherited finite cover; three new rows added.
S37	Fixed-norm trace-zero counts and unshifted quantitative sieve; extended to all shifts and odd degrees.
S14	Inherited all-norm cyclic prime-trace restrictions; used in the positive-density floor synthesis.
S20	Prior norm-one trace theorem in discriminant 1524; used only in the explicitly identified field corollary.
Heisei R05	A cyclic cubic seed for finite-sieve survival; the coset-counting theorem is proved here.
Old [007], [008]	Previously resolved dyadic Fibonacci and nonreversible recurrence problems.
Old [009], [031]	Earlier local obstructions, the three-adic embedding lemma, nonreal case, qualitative density and dimension results.

The other supplied topics were screened for useful connections. The pure-cubic sixth-power observation recovers results already subsumed by the old nonreal-cubic argument. No proved identity in the Gieseking, integral-period, or elliptic-resolvent notes supplies a missing prime-divisor theorem for a surviving Mills candidate. No such speculative connection is used as a proof input.

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[12] Showa, Heisei, and Taisho, *Constant proof race: Mills–Pisot submissions*, user-supplied research notes, 21 October 2026. In particular Showa rounds 12, 20, 27, 28, 35, 36, 37 and Heisei round 05. Source map above identifies the exact uses. Not external publications.

[13] Keio, *Dyadic Lucas Prime Cascades and Composite Fibonacci Shifts*, user-supplied manuscript [007], Theorem 1.1 and Corollary 1.2.

[14] Keio, *Additive Clocks, Non-Reversible Recurrences, and Transcendental Prime-Representing Constants*, user-supplied manuscript [008], Theorem 4.3.

[15] Keio, *Composite Terms in Sparse Cubic Pisot Floors (Alternating Shifts, 3-Adic Limits, and Projective Return)*, user-supplied manuscript [009].

[16] Keio, *Prime-Representing Constants and Related Structures (Encodings, Arithmetic, Geometry, and Modular Obstructions)*, user-supplied manuscript [031]. In particular Sections 14–15 and the earlier cubic-floor and density arguments.